In[•]:=
```
1.; 3^100
```

Out[•]=
```
515 377 520 732 011 331 036 461 129 765 621 272 702 107 522 001
```

In[•]:=
```
2.; 3^1000
```

Out[•]=
```
1 322 070 819 480 806 636 890 455 259 752 144 365 965 422 032 752 148 167 664 920 368 226 828 597 346 ⁞
 704 899 540 778 313 850 608 061 963 909 777 696 872 582 355 950 954 582 100 618 911 865 342 725 257 ⁞
 953 674 027 620 225 198 320 803 878 014 774 228 964 841 274 390 400 117 588 618 041 128 947 815 623 ⁞
 094 438 061 566 173 054 086 674 490 506 178 125 480 344 405 547 054 397 038 895 817 465 368 254 916 ⁞
 136 220 830 268 563 778 582 290 228 416 398 307 887 896 918 556 404 084 898 937 609 373 242 171 846 ⁞
 359 938 695 516 765 018 940 588 109 060 426 089 671 438 864 102 814 350 385 648 747 165 832 010 614 ⁞
 366 132 173 102 768 902 855 220 001
```

In[•]:=
```
3.; N[%]
```
数值运

Out[•]=
$$1.322070819480807 \times 10^{477}$$

In[•]:=
```
4.; pi = N[π, 200]
```
数值运算

Out[•]=
```
3.14159265358979323846264338327950288419716939937510582097494459230781640628620899862 ⁞
 03482534211706798214808651328230664709384460955058223172535940812848111745028410270 ⁞
 93852110555964462294895493038 20
```

In[•]:=
```
5.; E^((Sqrt[163] pi) / 3)
```
… 平方根

Out[•]=
```
640 320.000000000604863735049016039471741818818539475771485760366591819465221825828694 ⁞
 25363408158226464775899925470001727925679647867303997506923198495665261269396826561 ⁞
 38913713441824901972334433095 99
```

In[•]:=
```
Answer：小数点后有很多零;
```

In[•]:=
```
6.; StringPosition[StringDrop[ToString[N[Pi, 1000]], 2], "999999"]
```
字符串位置 删除指定… 转换为… … 圆周率

Out[•]=
```
{{762, 767}}
```

In[•]:=
```
7.; Sin[13 π]
```
正弦

Out[•]=
```
0
```

In[•]:=
```
Log[2.1]
```
对数

Out[•]=
```
0.741937
```

In[•]:=
```
E^(Sqrt[-1] π)
```
… 平方根

Out[•]=
$$-1$$

In[•]:=
```
8.; cosPiBy5 = Cos[π / 5]
```
余弦

Out[•]=
$$\frac{1}{4}\left(1+\sqrt{5}\right)$$

In[•]:=
```
closedForm = (1 + Sqrt[5]) / 4
```
平方根

```
cosPiBy5 == closedForm // FullSimplify
```
完全简化

Out[•]=
$$\frac{1}{4}\left(1+\sqrt{5}\right)$$

Out[•]=
```
True
```

In[•]:=
```
9.; FactorInteger[70 612 139 395 722 186]
```
整数因子分解

Out[•]=
```
{{2, 1}, {3, 2}, {43, 5}, {26 684 839, 1}}
```

In[•]:=
```
10.; Times @@ Apply[Power, {{2, 1}, {3, 2}, {43, 5}, {26 684 839, 1}}, 1]
```
乘 应用 幂

Out[•]=
```
70 612 139 395 722 186
```

In[•]:=
```
11.; (x^2 + 2 x + 1) ^20
```

Out[•]=
$$\left(1+2x+x^2\right)^{20}$$

In[•]:=
```
Expand[%]
```
展开

Out[•]=

$$1 + 40\,x + 780\,x^2 + 9880\,x^3 + 91\,390\,x^4 + 658\,008\,x^5 + 3\,838\,380\,x^6 + 18\,643\,560\,x^7 + 76\,904\,685\,x^8 + 273\,438\,880\,x^9 + 847\,660\,528\,x^{10} + 2\,311\,801\,440\,x^{11} + 5\,586\,853\,480\,x^{12} + 12\,033\,222\,880\,x^{13} + 23\,206\,929\,840\,x^{14} + 40\,225\,345\,056\,x^{15} + 62\,852\,101\,650\,x^{16} + 88\,732\,378\,800\,x^{17} + 113\,380\,261\,800\,x^{18} + 131\,282\,408\,400\,x^{19} + 137\,846\,528\,820\,x^{20} + 131\,282\,408\,400\,x^{21} + 113\,380\,261\,800\,x^{22} + 88\,732\,378\,800\,x^{23} + 62\,852\,101\,650\,x^{24} + 40\,225\,345\,056\,x^{25} + 23\,206\,929\,840\,x^{26} + 12\,033\,222\,880\,x^{27} + 5\,586\,853\,480\,x^{28} + 2\,311\,801\,440\,x^{29} + 847\,660\,528\,x^{30} + 273\,438\,880\,x^{31} + 76\,904\,685\,x^{32} + 18\,643\,560\,x^{33} + 3\,838\,380\,x^{34} + 658\,008\,x^{35} + 91\,390\,x^{36} + 9880\,x^{37} + 780\,x^{38} + 40\,x^{39} + x^{40}$$

In[•]:=
```
Factor[%]
```
因式分解

Out[•]=

$$(1 + x)^{40}$$

In[•]:=
```
12.; (x^2 + 2 x + 1) ^20
```

Out[•]=

$$\left(1 + 2\,x + x^2\right)^{20}$$

In[•]:=
```
Expand[(1 + 2 x + x^2) ^20]
```
展开

Out[•]=

$$1 + 40\,x + 780\,x^2 + 9880\,x^3 + 91\,390\,x^4 + 658\,008\,x^5 + 3\,838\,380\,x^6 + 18\,643\,560\,x^7 + 76\,904\,685\,x^8 + 273\,438\,880\,x^9 + 847\,660\,528\,x^{10} + 2\,311\,801\,440\,x^{11} + 5\,586\,853\,480\,x^{12} + 12\,033\,222\,880\,x^{13} + 23\,206\,929\,840\,x^{14} + 40\,225\,345\,056\,x^{15} + 62\,852\,101\,650\,x^{16} + 88\,732\,378\,800\,x^{17} + 113\,380\,261\,800\,x^{18} + 131\,282\,408\,400\,x^{19} + 137\,846\,528\,820\,x^{20} + 131\,282\,408\,400\,x^{21} + 113\,380\,261\,800\,x^{22} + 88\,732\,378\,800\,x^{23} + 62\,852\,101\,650\,x^{24} + 40\,225\,345\,056\,x^{25} + 23\,206\,929\,840\,x^{26} + 12\,033\,222\,880\,x^{27} + 5\,586\,853\,480\,x^{28} + 2\,311\,801\,440\,x^{29} + 847\,660\,528\,x^{30} + 273\,438\,880\,x^{31} + 76\,904\,685\,x^{32} + 18\,643\,560\,x^{33} + 3\,838\,380\,x^{34} + 658\,008\,x^{35} + 91\,390\,x^{36} + 9880\,x^{37} + 780\,x^{38} + 40\,x^{39} + x^{40}$$

In[•]:=
```
Factor[1 + 40 x + 780 x^2 + 9880 x^3 + 91 390 x^4 + 658 008 x^5 + 3 838 380 x^6 + 18 643 560 x^7 +
  76 904 685 x^8 + 273 438 880 x^9 + 847 660 528 x^10 + 2 311 801 440 x^11 + 5 586 853 480 x^12 +
  12 033 222 880 x^13 + 23 206 929 840 x^14 + 40 225 345 056 x^15 + 62 852 101 650 x^16 +
  88 732 378 800 x^17 + 113 380 261 800 x^18 + 131 282 408 400 x^19 + 137 846 528 820 x^20 +
  131 282 408 400 x^21 + 113 380 261 800 x^22 + 88 732 378 800 x^23 + 62 852 101 650 x^24 +
  40 225 345 056 x^25 + 23 206 929 840 x^26 + 12 033 222 880 x^27 + 5 586 853 480 x^28 +
  2 311 801 440 x^29 + 847 660 528 x^30 + 273 438 880 x^31 + 76 904 685 x^32 + 18 643 560 x^33 +
  3 838 380 x^34 + 658 008 x^35 + 91 390 x^36 + 9880 x^37 + 780 x^38 + 40 x^39 + x^40]
```
因式分解

Out[•]=

$$(1 + x)^{40}$$

In[•]:=
```
13.; TrigReduce[Sin[x] Cos[y] - Cos[x] Sin[y]]
```
三角函数约化 正弦 余弦 余弦 正弦

Out[•]=

$$\text{Sin}[x - y]$$

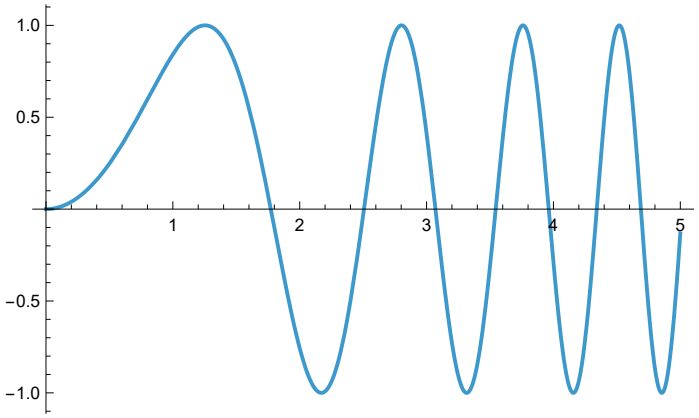
In[•]:=
```
TrigExpand[%]
```
三角函数展开

Out[•]=
```
Cos[y] Sin[x] - Cos[x] Sin[y]
```

In[•]:= **Answer：它们是互相的逆运算；**

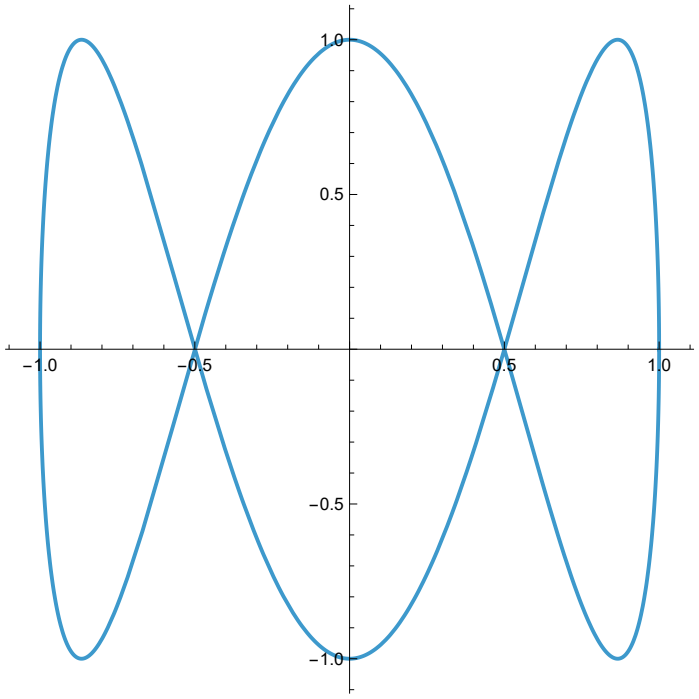
In[•]:=
```
14.; Plot[Sin[x^2], {x, 0, 5}]
```
绘图 正弦

Out[•]=



In[•]:=
```
15.; ParametricPlot[{Cos[x], Sin[3 x]}, {x, 0, 2 π}]
```
绘制参数图 余弦 正弦

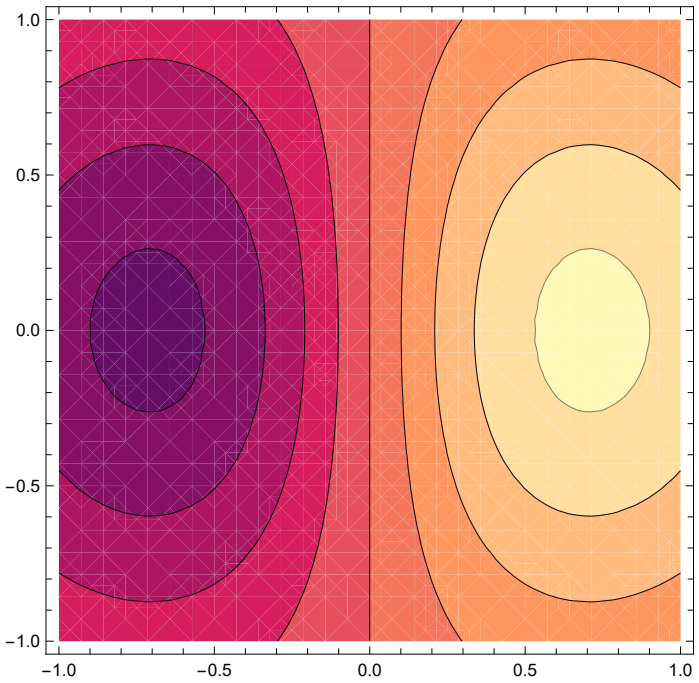
Out[•]=



In[•]:= **Answer：这是一个三叶型曲线；**

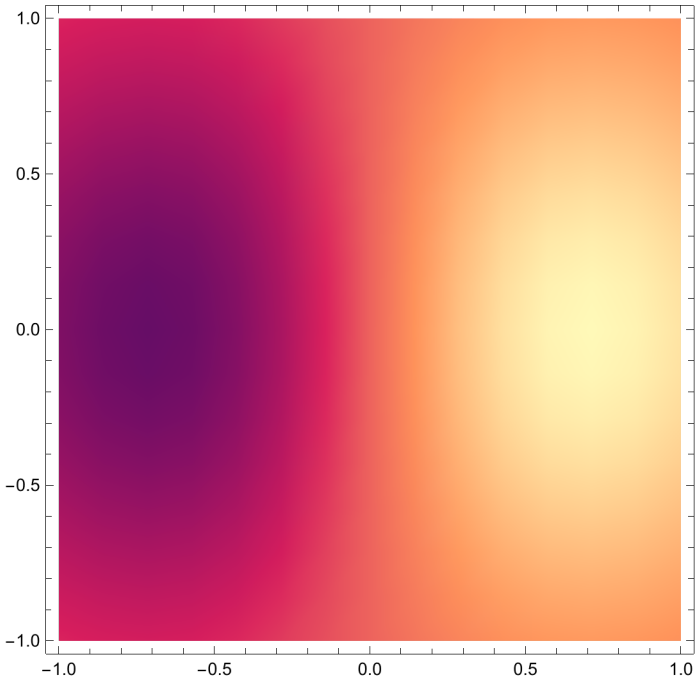
In[•]:=
```
16.; ContourPlot[x E^(-x^2 - y^2), {x, -1, 1}, {y, -1, 1}]
```
绘制等高线　　自然常数

Out[•]=



In[•]:=
```
DensityPlot[x E^(-x^2 - y^2), {x, -1, 1}, {y, -1, 1}]
```
密度图　　自然常数

Out[•]=



In[•]:= **17; $\int$ x / (x^3 – 1) dx**

Out[•]=

$$\frac{\text{ArcTan}\left[\frac{1+2x}{\sqrt{3}}\right]}{\sqrt{3}} + \frac{1}{3}\text{Log}[1-x] - \frac{1}{6}\text{Log}\left[1+x+x^2\right]$$

In[•]:= **∂_x (ArcTan[(1+2 x)/√3] / √3 + 1/3 Log[1 – x] – 1/6 Log[1 + x + x^2])**
对数 对数

Out[•]=

$$-\frac{1}{3\,(1-x)} - \frac{1+2x}{6\left(1+x+x^2\right)} + \frac{2}{3\left(1+\frac{1}{3}\,(1+2x)^2\right)}$$

In[•]:= **Simplify[%]**
化简

Out[•]=

$$\frac{x}{-1+x^3}$$

In[1]:= **18; $\int_0^\infty$ Sin^2[x] Cos^3[x] / x dx**

Out[1]= $$\int_0^\infty \frac{\text{Cos}^3[x]\,\text{Sin}^2[x]}{x}\,dx$$

In[3]:=
```
f[x_] = Sin[x]^2 Cos[x]^3 // TrigReduce
        正弦     余弦      三角函数约化
analytic = 1 / 16 (2 π / 2 + π / 2 - π / 2);
N[analytic, 20]
数值运算
numeric1 = NIntegrate[(Sin[x]^2 Cos[x]^3) / x,
           数值积分    正弦     余弦
  {x, 0, ∞}, Method → {"DoubleExponential", "SymbolicProcessing" → 0},
             方法
  WorkingPrecision → 20, MaxRecursion → 20]
  工作精度              最大递推
numeric2 = NIntegrate[(Sin[x]^2 Cos[x]^3) / x, {x, 0, ∞},
           数值积分    正弦     余弦
  Method → {"LevinRule"}, WorkingPrecision → 20, MaxRecursion → 20]
  方法                    工作精度               最大递推


Print["解析值 π/16 = ", N[analytic, 20]];
打印                   数值运算
Print["数值结果 (DoubleExponential)  = ", numeric1];
打印
Print["数值结果 (LevinRule)  = ", numeric2];
打印
Print["差值 (DoubleExponential)  = ", Abs[numeric1 - N[analytic, 20]]];
打印                                  绝对值      数值运算
Print["差值 (LevinRule)  = ", Abs[numeric2 - N[analytic, 20]]];
打印                          绝对值      数值运算
```

Out[3]=
$$\frac{1}{16}\left(2\cos[x]-\cos[3x]-\cos[5x]\right)$$

Out[5]=
```
0.19634954084936207740
```

In[16]:=
$$19.;\int_0^1 \text{Sin}[\text{Sin}[x]]\,dx$$
正弦 正弦

In[18]:=
```
NIntegrate[Sin[Sin[x]], {x, 0, 1}]
数值积分   正弦 正弦
```

Out[18]=
```
0.430606
```

In[19]:=
```
20.; s = Solve[a x^2 + b x + c == 0, x]
```
解方程

Out[19]=
$$\left\{\left\{x \to \frac{-b - \sqrt{b^2 - 4\,a\,c}}{2\,a}\right\}, \left\{x \to \frac{-b + \sqrt{b^2 - 4\,a\,c}}{2\,a}\right\}\right\}$$

In[20]:=
```
a x^2 + b x + c == 0 /. s
```

Out[20]=
$$\left\{c + \frac{b\left(-b - \sqrt{b^2 - 4\,a\,c}\right)}{2\,a} + \frac{\left(-b - \sqrt{b^2 - 4\,a\,c}\right)^2}{4\,a} == 0,\ c + \frac{b\left(-b + \sqrt{b^2 - 4\,a\,c}\right)}{2\,a} + \frac{\left(-b + \sqrt{b^2 - 4\,a\,c}\right)^2}{4\,a} == 0\right\}$$

In[21]:=
```
Simplify[%]
```
化简

Out[21]=
```
{True, True}
```

In[22]:=
```
21.; Solve[x^3 + x + 1 == 0, x]
```
解方程

Out[22]=
$$\left\{\left\{x \to \sqrt[3]{-0.682\ldots}\right\}, \left\{x \to \sqrt[3]{0.341\ldots - 1.16\ldots\,i}\right\}, \left\{x \to \sqrt[3]{0.341\ldots + 1.16\ldots\,i}\right\}\right\}$$

In[23]:=
```
N[%]
```
数值运算

Out[23]=
```
{{x → -0.682328}, {x → 0.341164 - 1.16154 i}, {x → 0.341164 + 1.16154 i}}
```

In[24]:=
```
Answer : My answer satisfy the requirement;
```

In[25]:=
```
22.; mat = {{3.5, 7.2}, {-2.4, 6.4}}
```

Out[25]=
```
{{3.5, 7.2}, {-2.4, 6.4}}
```

In[26]:=
```
Inverse[mat]
```
逆
```
Eigenvalues[mat]
```
特征值

Out[26]=
```
{{0.16129, -0.181452}, {0.0604839, 0.0882056}}
```

Out[27]=
```
{4.95 + 3.89583 i, 4.95 - 3.89583 i}
```

In[28]:=
```
23.;
mat = {{a, b, c}, {d, e, f}, {g, h, f}}
```

Out[29]=
$$\{\{a, b, c\}, \{d, e, f\}, \{g, h, f\}\}$$

In[30]:=
```
Inverse[mat]
```
逆

Out[30]=
$$\left\{\left\{\frac{e f - f h}{-b d f + a e f - c e g + b f g + c d h - a f h}, \frac{-b f + c h}{-b d f + a e f - c e g + b f g + c d h - a f h}, \frac{-c e + b f}{-b d f + a e f - c e g + b f g + c d h - a f h}\right\}, \left\{\frac{-d f + f g}{-b d f + a e f - c e g + b f g + c d h - a f h}, \frac{a f - c g}{-b d f + a e f - c e g + b f g + c d h - a f h}, \frac{c d - a f}{-b d f + a e f - c e g + b f g + c d h - a f h}\right\}, \left\{\frac{-e g + d h}{-b d f + a e f - c e g + b f g + c d h - a f h}, \frac{b g - a h}{-b d f + a e f - c e g + b f g + c d h - a f h}, \frac{-b d + a e}{-b d f + a e f - c e g + b f g + c d h - a f h}\right\}\right\}$$

In[31]:=
```
minv = %
```

Out[31]=
$$\left\{\left\{\frac{e f - f h}{-b d f + a e f - c e g + b f g + c d h - a f h}, \frac{-b f + c h}{-b d f + a e f - c e g + b f g + c d h - a f h}, \frac{-c e + b f}{-b d f + a e f - c e g + b f g + c d h - a f h}\right\}, \left\{\frac{-d f + f g}{-b d f + a e f - c e g + b f g + c d h - a f h}, \frac{a f - c g}{-b d f + a e f - c e g + b f g + c d h - a f h}, \frac{c d - a f}{-b d f + a e f - c e g + b f g + c d h - a f h}\right\}, \left\{\frac{-e g + d h}{-b d f + a e f - c e g + b f g + c d h - a f h}, \frac{b g - a h}{-b d f + a e f - c e g + b f g + c d h - a f h}, \frac{-b d + a e}{-b d f + a e f - c e g + b f g + c d h - a f h}\right\}\right\}$$

In[32]:= **minv.mat**

Out[32]=

$$\left\{\left\{\frac{(-ce+bf)\,g}{-bdf+aef-ceg+bfg+cdh-afh}+\frac{d\,(-bf+ch)}{-bdf+aef-ceg+bfg+cdh-afh}+\frac{a\,(ef-fh)}{-bdf+aef-ceg+bfg+cdh-afh},\frac{(-ce+bf)\,h}{-bdf+aef-ceg+bfg+cdh-afh}+\frac{e\,(-bf+ch)}{-bdf+aef-ceg+bfg+cdh-afh}+\frac{b\,(ef-fh)}{-bdf+aef-ceg+bfg+cdh-afh},\frac{f\,(-ce+bf)}{-bdf+aef-ceg+bfg+cdh-afh}+\frac{f\,(-bf+ch)}{-bdf+aef-ceg+bfg+cdh-afh}+\frac{c\,(ef-fh)}{-bdf+aef-ceg+bfg+cdh-afh}\right\},\right.$$

$$\left\{\frac{(cd-af)\,g}{-bdf+aef-ceg+bfg+cdh-afh}+\frac{d\,(af-cg)}{-bdf+aef-ceg+bfg+cdh-afh}+\frac{a\,(-df+fg)}{-bdf+aef-ceg+bfg+cdh-afh},\frac{e\,(af-cg)}{-bdf+aef-ceg+bfg+cdh-afh}+\frac{b\,(-df+fg)}{-bdf+aef-ceg+bfg+cdh-afh}+\frac{(cd-af)\,h}{-bdf+aef-ceg+bfg+cdh-afh},\frac{f\,(cd-af)}{-bdf+aef-ceg+bfg+cdh-afh}+\frac{f\,(af-cg)}{-bdf+aef-ceg+bfg+cdh-afh}+\frac{c\,(-df+fg)}{-bdf+aef-ceg+bfg+cdh-afh}\right\},$$

$$\left.\left\{\frac{(-bd+ae)\,g}{-bdf+aef-ceg+bfg+cdh-afh}+\frac{d\,(bg-ah)}{-bdf+aef-ceg+bfg+cdh-afh}+\frac{a\,(-eg+dh)}{-bdf+aef-ceg+bfg+cdh-afh},\frac{(-bd+ae)\,h}{-bdf+aef-ceg+bfg+cdh-afh}+\frac{e\,(bg-ah)}{-bdf+aef-ceg+bfg+cdh-afh}+\frac{b\,(-eg+dh)}{-bdf+aef-ceg+bfg+cdh-afh},\frac{(-bd+ae)\,f}{-bdf+aef-ceg+bfg+cdh-afh}+\frac{f\,(bg-ah)}{-bdf+aef-ceg+bfg+cdh-afh}+\frac{c\,(-eg+dh)}{-bdf+aef-ceg+bfg+cdh-afh}\right\}\right\}$$

In[33]:= **Simplify[%]**
化简

Out[33]=

```
{{1, 0, 0}, {0, 1, 0}, {0, 0, 1}}
```

In[34]:= **24; rand = Table[RandomReal[1, {3, 3}]]**
表格 伪随机实数

Out[34]=

```
{{0.127776, 0.277487, 0.490566},
 {0.596675, 0.446468, 0.888254}, {0.60003, 0.373059, 0.686754}}
```

In[35]:=
```
rand[[2, 2]] = x
```

Out[35]=
```
x
```

In[36]:=
```
inv = Inverse[rand]
```
逆

Out[36]=
$$\left\{\left\{\frac{-0.331371+0.686754\,x}{0.101045-0.206603\,x},\ -\frac{0.00755539}{0.101045-0.206603\,x},\ \frac{0.246479-0.490566\,x}{0.101045-0.206603\,x}\right\},\right.$$
$$\left\{\frac{0.12321}{0.101045-0.206603\,x},\ -\frac{0.206603}{0.101045-0.206603\,x},\ \frac{0.179211}{0.101045-0.206603\,x}\right\},$$
$$\left.\left\{\frac{0.222595-0.60003\,x}{0.101045-0.206603\,x},\ \frac{0.118832}{0.101045-0.206603\,x},\ \frac{-0.16557+0.127776\,x}{0.101045-0.206603\,x}\right\}\right\}$$

In[37]:=
```
inv /. x → 1
```

Out[37]=
```
{{-3.36671, 0.0715756, 2.31235},
 {-1.16722, 1.95725, -1.69774}, {3.57562, -1.12575, 0.358031}}
```

In[38]:=
```
25.; is = Table[i!, {i, 1, 10}]
```
表格

Out[38]=
```
{1, 2, 6, 24, 120, 720, 5040, 40 320, 362 880, 3 628 800}
```

In[39]:=
```
26.; Log[is]
```
对数

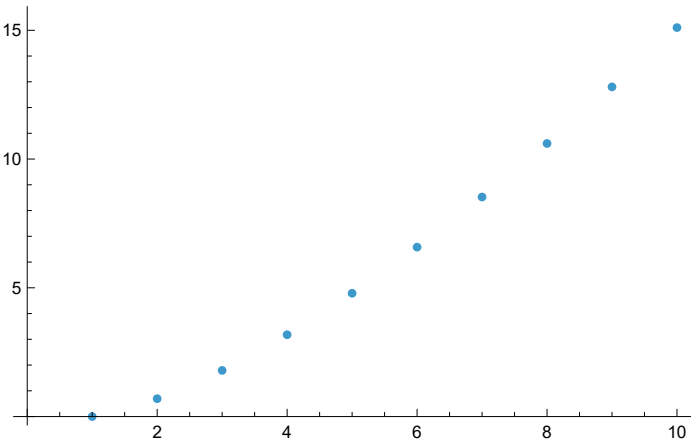
Out[39]=
```
{0, Log[2], Log[6], Log[24], Log[120], Log[720],
 Log[5040], Log[40 320], Log[362 880], Log[3 628 800]}
```

In[40]:=
```
data = N[%]
```
数值运

Out[40]=
```
{0., 0.693147, 1.79176, 3.17805, 4.78749, 6.57925, 8.52516, 10.6046, 12.8018, 15.1044}
```

In[41]:=
```
27; p1 = ListPlot[data]
```
绘制点集

Out[41]=



In[42]:=
```
28.; data = Table[{i, N[Log[i!]]}, {i, 1, 10}]
```
表格 对数

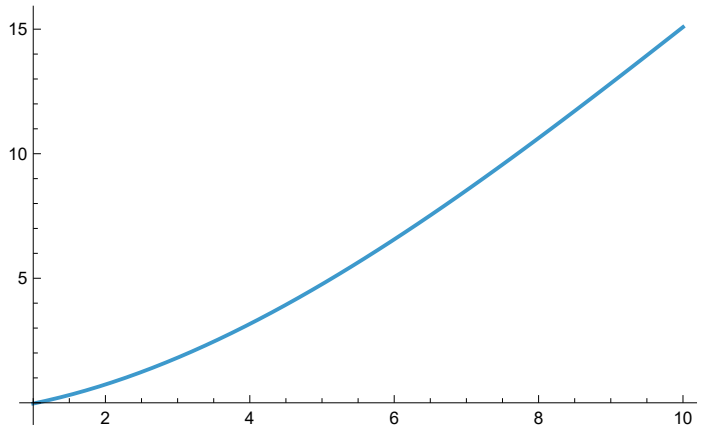
Out[42]=
```
{{1, 0.}, {2, 0.693147}, {3, 1.79176}, {4, 3.17805}, {5, 4.78749},
 {6, 6.57925}, {7, 8.52516}, {8, 10.6046}, {9, 12.8018}, {10, 15.1044}}
```

In[43]:=
```
fit = Fit[data, {1, x, x^2, x^3}, x]
```
拟合

Out[43]=
$$-0.445615 + 0.229907\,x + 0.192215\,x^2 - 0.00599207\,x^3$$

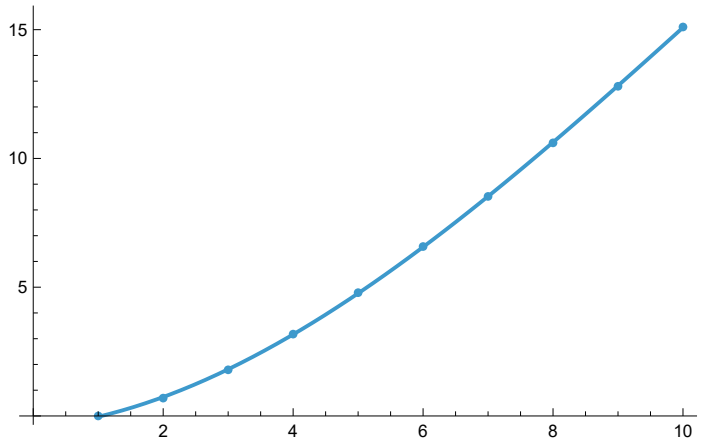
In[44]:=
```
p2 = Plot[fit, {x, 1, 10}, PlotRange → All]
```
绘图 绘制范围 全部

Out[44]=



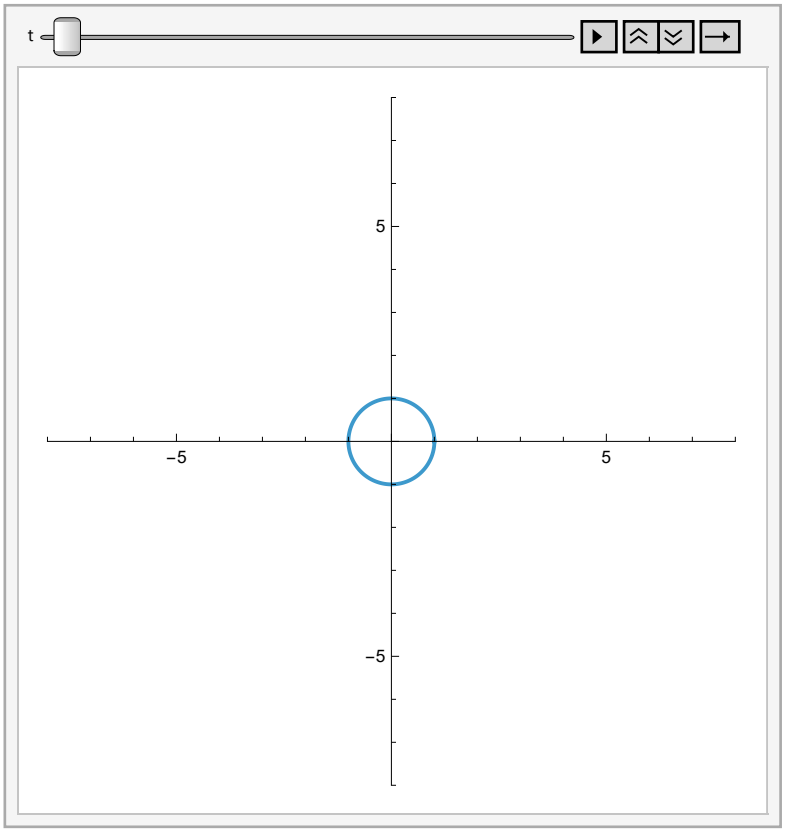
In[45]:=
```
Show[p1, p2]
```
显示

Out[45]=



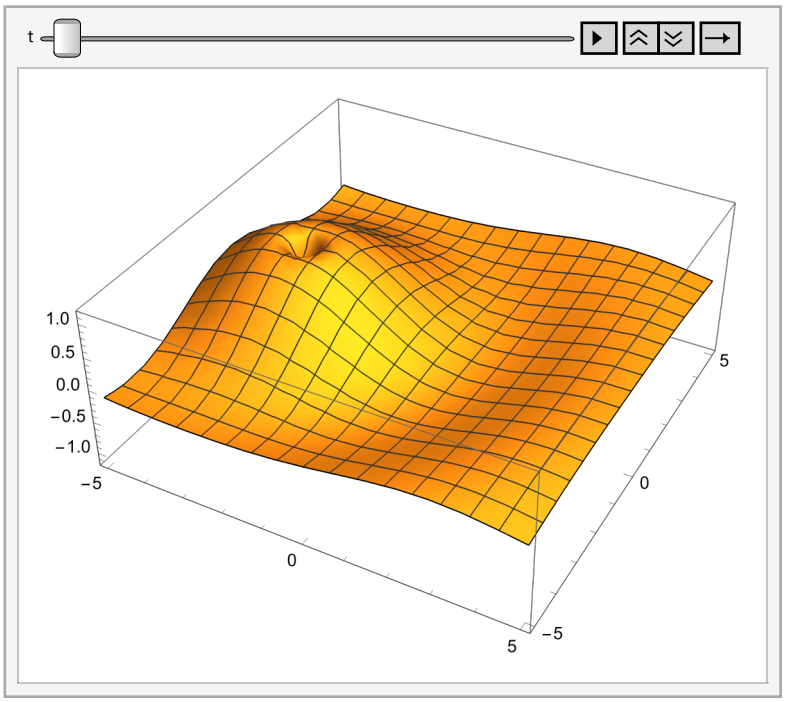
In[50]:=
```
29.;
ani2D = Animate[ParametricPlot[{t Cos[s], t Sin[s]}, {s, 0, 2 π},
        生成动画  绘制参数图              余弦       正弦
    PlotRange → {{-8, 8}, {-8, 8}}], {t, 1, 6}, AnimationRunning → False]
    绘制范围                                    动画播放状态          假
```

Out[50]=



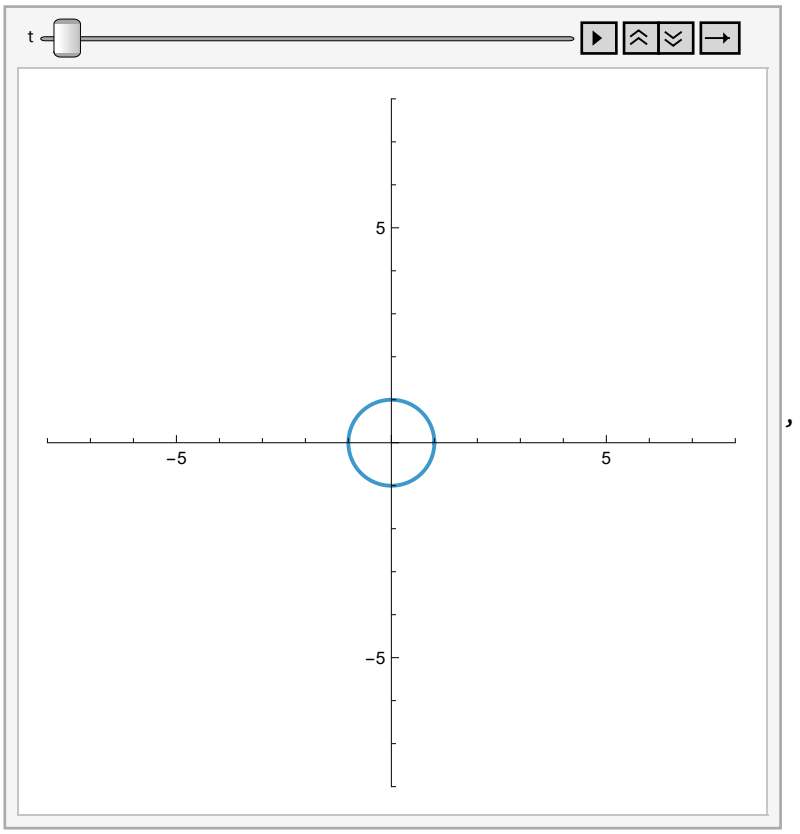
In[51]:=
```
ani3D = Animate[Plot3D[Sin[Sqrt[(x - t)^2 + y^2]] / Sqrt[(x - t)^2 + y^2 + 0.1], {x, -5, 5},
                生成动画  绘制…  正弦 平方根                    平方根
  {y, -5, 5}, PlotRange → {-1.2, 1.2}], {t, -3, 3}, AnimationRunning → False]
              绘制范围                              动画播放状态          假
```

Out[51]=

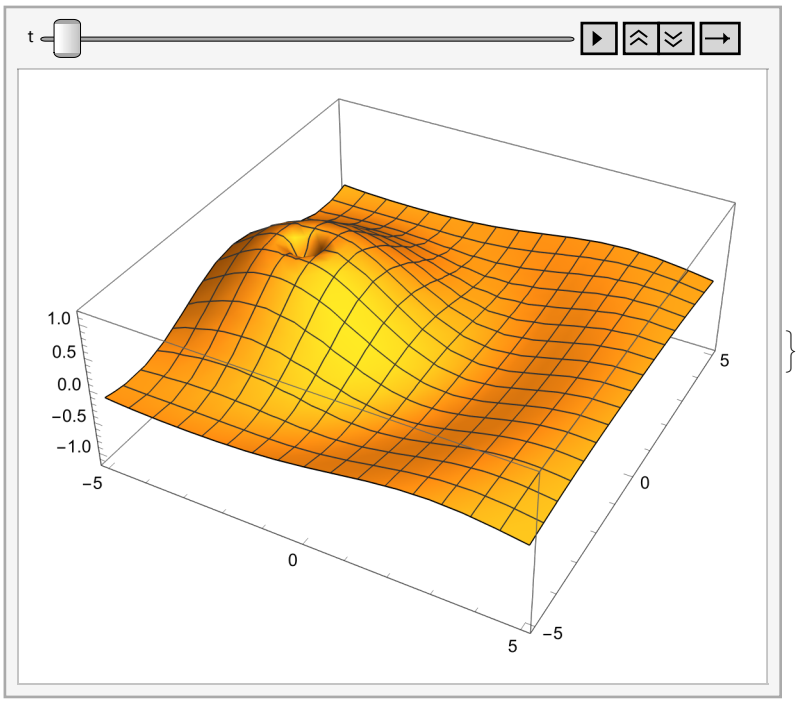


In[52]:= **{ani2D, ani3D}**

Out[52]=



,



In[54]:=
```
30.; Image@Array[Mod[# + #2, 2] &, {64, 64}]
```
图像 数组 模余

Out[54]=

